Function Machines

Year 4 Pt 2 · **Mark Scheme**

Lesson Worksheet — Answers

Function Basics

1. 30

2. 23

3. 14

4. C) 8

5. 18 (4 + 2 = 6; 6 × 3 = 18)

6. D) 23 (9 × 2 = 18; 18 + 5 = 23)

7. 7 (42 ÷ 6 = 7)

8. C) 16 (31 − 15 = 16)

9. 13 (5 + 8 = 13)

10. 45 (9 × 5 = 45)

11. A, B and D (4×3=12, 5×3=15, 7×3=21 are correct; C is false since 6×3=18 not 20, and E is false since 8×3=24 not 22)

12. 17 (5 × 4 = 20; 20 − 3 = 17)

13. 8 (9 + 7 = 16; 16 ÷ 2 = 8)

14. D) 26 (10 × 2 = 20; 20 + 6 = 26)

15. 0

Working Forwards & Backwards

16. 41 (11 × 3 = 33; 33 + 8 = 41)

17. C) 28 (12 − 5 = 7; 7 × 4 = 28)

18. 4 (18 ÷ 3 = 6; 6 − 2 = 4)

19. C) 7 (33 ÷ 3 = 11; 11 − 4 = 7)

20. 12 (15 + 9 = 24; 24 ÷ 2 = 12)

21. a = 3, b = 4 (so the rule is ×3, then +4): from 5a+b=19 and 8a+b=28, subtracting gives 3a=9 so a=3, then b=19−15=4; check 8×3+4=28 ✓

22. 2 (0.5 × 4 = 2)

23. C) 8 (29 − 5 = 24; 24 ÷ 3 = 8)

24. 4 (−3 + 7 = 4)

25. A, C and E (6→4, 10→12 and 15→22 are correct; B is false since (9−4)×2=10 not 11, and D is false since (12−4)×2=16 not 14)

26. 2 (2/5 × 5 = 2)

27. 7 (10 + 4 = 14; 14 ÷ 2 = 7)

28. C) 6.5 (13 × 0.5 = 6.5)

29. 4 (−8 ÷ 4 = −2; −2 + 6 = 4)

30. 2 (3/4 + 1/4 = 1; 1 × 2 = 2)

Multi-Step Challenge

31. 11 (10 × 4 = 40; 40 − 7 = 33; 33 ÷ 3 = 11)

32. 5 (−12 ÷ 3 = −4; −4 + 9 = 5)

33. C) 12 (6a+b=32 and 9a+b=44 give 3a=12 so a=4, then b=32−24=8; a+b=12)

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34. 26 (9 − 2.5 = 6.5; 6.5 × 4 = 26)

35. −3 (6 × −2 = −12; −12 + 9 = −3)

36. A, B and C (5→3, 8→9 and 10→13 are correct since 2x−7 gives these values; D is false since 2×12−7=17 not 18, and E is false since 2×20−7=33 not 35)

37. C) 15 (12 ÷ 2 = 6; 6 × 5 = 30; 30 − 15 = 15)

38. 3 (1 + 20 = 21; 21 ÷ 7 = 3)

39. −15 (4 + 11 = 15; 15 × −1 = −15)

40. 1 (9 ÷ 2 = 4.5; 4.5 − 3.5 = 1)

41. C) 10 (2x+9=29 since double of the answer is 29; 2x=20 so x=10)

42. £29 (9 × 3 = 27; 27 + 2 = 29)

43. 2 (−2 × 3 = −6; −6 + 8 = 2)

44. C) 17 (5 × 6 = 30; 30 + 4 = 34; 34 ÷ 2 = 17)

45. 9 (6 × 5/3 = 10; 10 − 1 = 9)

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Homework Worksheet — Answers

Function Basics

1. 56

2. 37

3. 21

4. B) 7

5. 22 (6 + 5 = 11; 11 × 2 = 22)

6. C) 28 (8 × 3 = 24; 24 + 4 = 28)

7. 6 (54 ÷ 9 = 6)

8. C) 22 (40 − 18 = 22)

9. 20 (9 + 11 = 20)

10. 42 (6 × 7 = 42)

11. A, C and D (3→12, 6→24 and 7→28 are correct; B is false since 5×4=20 not 18, and E is false since 9×4=36 not 32)

12. 14 (4 × 5 = 20; 20 − 6 = 14)

13. 7 (12 + 9 = 21; 21 ÷ 3 = 7)

14. C) 13 (6 × 3 = 18; 18 − 5 = 13)

15. 0

Applied Functions

16. 29 (9 × 2 = 18; 18 + 11 = 29)

17. B) 40 (15 − 7 = 8; 8 × 5 = 40)

18. 15 (24 ÷ 4 = 6; 6 + 9 = 15)

19. B) 8 (28 ÷ 2 = 14; 14 − 6 = 8)

20. 7 (11 + 10 = 21; 21 ÷ 3 = 7)

21. a = 3, b = 3 (so the rule is ×3, then +3): from 4a+b=15 and 7a+b=24, subtracting gives 3a=9 so a=3, then b=15−12=3; check 7×3+3=24 ✓

22. 3 (0.5 × 6 = 3)

23. C) 8 (23 − 7 = 16; 16 ÷ 2 = 8)

24. 4 (−6 + 10 = 4)

25. A, C and E (2→15, 5→24 and 7→30 are correct; B is false since (4+3)×3=21 not 20, and D is false since (6+3)×3=27 not 28)

26. 3 (3/4 × 4 = 3)

27. C) 9.5 (19 × 0.5 = 9.5)

28. 8 (18 + 6 = 24; 24 ÷ 3 = 8)

29. 6 (−10 ÷ 5 = −2; −2 + 8 = 6)

30. 3 (2/3 + 1/3 = 1; 1 × 3 = 3)

Standard / Advanced

31. 7 (20 − 6 = 14; 14 ÷ 2 = 7)

32. 8 (15 + 9 = 24; 24 ÷ 3 = 8)

33. 11 (24 ÷ 4 = 6; 6 + 5 = 11)

34. 3 hours (£90 − £15 = £75; £75 ÷ £25 = 3)

35. 11 (30 ÷ 2 = 15; 15 − 4 = 11)

36. 20 ((2 + 8) × 3 = 30; 30 − 10 = 20)

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37. C) 9 (£24 − £6 = £18; £18 ÷ £2 = 9 rides)

38. 34 ((5 × 2) + 7 = 17; 17 × 2 = 34)

39. C) 32 (from 3a+b=11 and 6a+b=20, 3a=9 so a=3, b=2; 10×3+2=32)

40. 20 (16 − 9 = 7; 7 × 2 = 14; 14 + 6 = 20)

41. A, C and E (2→11, 6→19 and 10→27 are correct since 2(x+5)−3 gives these values; B is false since the true value is 15 not 16, and D is false since the true value is 23 not 24)

42. 10 km (£11.50 − £3.50 = £8; £8 ÷ £0.80 = 10)

43. 23 (9 × 2 = 18; 18 + 5 = 23)

44. B) 25 (5 + 9 = 14; 14 × 2 = 28; 28 − 3 = 25)

45. 7 (first pass: 4×2−3=5; second pass: 5×2−3=7)

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